Beyond Calibration:
Physics Guidance for Scientific Coding Agents

Project 1 Proposal · September 18, 2026

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Evaluation goal and claim. A recurring failure in our scientific-agent evaluations is getting the physics wrong despite fitting calibration data well and passing public tests. We will develop PhysLiteBench to test whether physics hints help agents correct these errors more effectively than generic correctness reminders. Existing evaluations study scientific task success and expert guidance [1,2]; we isolate *physical model selection* using correct and shortcut models that agree on calibration but diverge on automatically scored, unseen preparations. The October 16 deliverable is a reproducible benchmark and results for three Luna configurations: no hint, a generic reminder, and physics guidance.

Task and episode. We start with three PhysLiteBench systems: piston-coupled gases, spins in a shared fluctuating field, and coupled salt diffusion. In a fresh Docker container, Luna (`gpt-5.6-luna`, high reasoning effort) receives apparatus documentation, noisy calibration data, public tests, and a Python starter. It has shell/Python tools and 10 minutes to edit `model.py`, fit one parameter, and predict new preparations. Reference solutions and hidden tests remain unavailable until verification.

Automatic metric. An episode scores 1 only if every public and private test passes; otherwise it scores 0. Current tasks require reduced calibration $\chi^2 < 1.5$, relative parameter error below 5%, and hidden prediction errors below fixed thresholds: probability RMSE < 0.03 (spin) or normalized RMSE < 5% (thermal/diffusion). New tasks will have thresholds validated against independent references and completed shortcuts before agent runs. Report per-task pass rates, differences between configurations, and binomial confidence intervals. Scoring is automatic; trajectory review diagnoses failures without changing scores.

Baselines and preliminary evidence. Compare identical agents with no added hint, a generic correctness reminder, or a physics hint; hold data, tools, grading, and limits fixed. The 15-trial thermal pilot, with five trials per condition, found:

Instruction	No hint	Generic reminder	Physics hint
Full passes	0/5	1/5	5/5
Calibration and parameter checks	5/5	5/5	5/5

The physics hint asks the agent to apply the first law of thermodynamics separately to each gas, accounting for heat flow and work exchanged as the piston moves. Mechanical equilibrium makes the chamber pressures equal at each instant, but their common pressure can still change as the gases exchange heat with the bath. The starter incorrectly extends a constant-pressure approximation beyond the calibration conditions, where it happens to agree with the correct model. The hint therefore targets an error in the governing equations that calibration fit alone cannot reveal. These preliminary results motivate testing the same comparison across the expanded task suite.

By October 2. Design and implement **ten tasks in total**, including the existing three, with independent physical references, calibration-fitting shortcuts, and automatic hidden tests. Complete initial runs of all three configurations and **trajectory analysis**: inspect recorded explanations, tool calls, code revisions, and test results to locate where agents misinterpret physical assumptions, derive incorrect equations, or abandon a correct candidate. Distinguish these failures from mathematical, numerical, coding, and infrastructure errors. Deliver the runnable suite, archived trajectories, and initial analysis. Freeze prompts and graders before each batch, match reminder lengths, randomize execution order, and keep the pilot separate.

Main risk. Hints may supply the decisive task-specific insight without improving general reasoning. Use guidance on physical principles, evaluate across ten tasks, and limit conclusions to the tested systems and model. Repeated runs measure sampling variability, not physical diversity.

Compute and access. Available resources are personal ChatGPT/Codex subscription access and local CPU/Docker; no API credits or institutional compute are committed. The October 16 target is 10 tasks × 3 configurations × 10 fresh trials: 300 episodes (at most 50 agent-hours), plus CPU verification. No GPU or training is needed; if quotas bind, reduce repeats uniformly and report actual counts.

References. [1] Chen et al., *ScienceAgentBench*, ICLR 2025. [2] Xu et al., *SWE-bench Science*, arXiv, 2026.